

TABLE I
DATASET COMPOSITION AND LEAKAGE-CONTROLLED EVALUATION PROTOCOL

Dataset / Partition	Images	Classes	Purpose
PlantVillage Raw	16,011	10	Initial tomato disease dataset
PlantVillage Clean	15,997	10	After exact-duplicate removal
Training	11,199	10	Model fitting
Validation	2,399	10	Model selection
Internal Test	2,399	10	Independent internal evaluation
PlantDoc External Test	79	8 mapped	Independent zero-shot evaluation

Leakage audit: 14 exact duplicates removed; cross-split exact/pHash leakage = 0; PlantVillage-PlantDoc exact/near-duplicate overlap = 0 (locked pHash threshold ≤ 4).